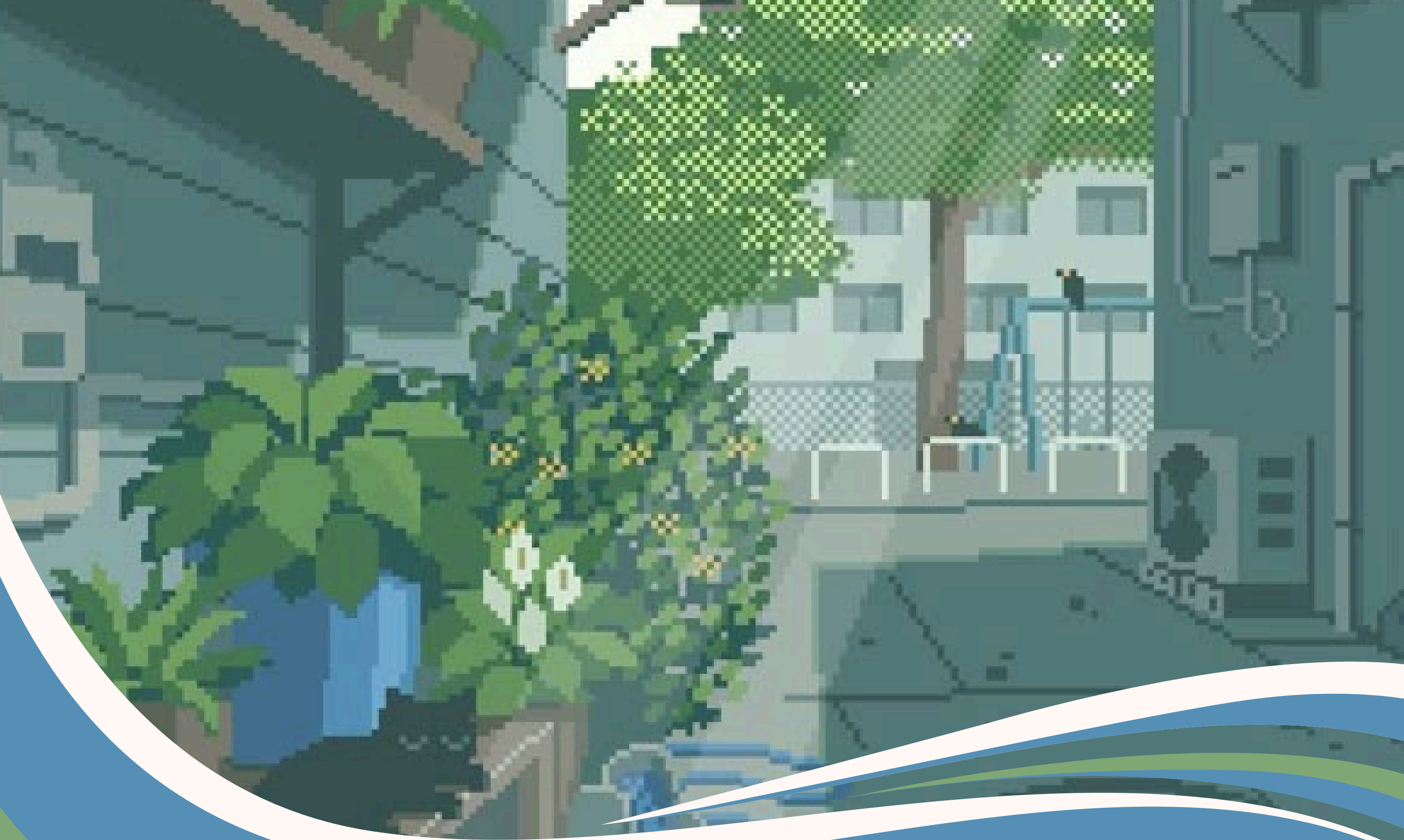

• 2025 •

TECHNICAL INSTALLATION MANUAL

GAMIFIED FINANCIAL VISUALIZER

For

Demo 4

Presented by :

CodeBlooded

MENU

Introduction

This document provides a comprehensive guide for setting up and deploying the Gamified Finance application—a multi-service system designed to deliver personalized financial insights through engaging, game-like interactions. It outlines the required hardware and software components, and walks through the installation process for each service, including the React-based frontend, Node.js backend, and Python-powered AI module.

Whether you're a developer contributing to the project or a stakeholder reviewing the system architecture, this manual ensures a smooth setup experience across major operating systems (Windows, Linux, macOS), with emphasis on cloud deployment via Azure and CI/CD integration.

Shells:
Windows → PowerShell (or Command Prompt).
macOS/Linux → Terminal (bash/zsh).
Commands marked "(PowerShell)" have Windows-specific activation syntax.

Prerequisites

Computer Hardware Requirements

Hardware	Minimum	Recommended
CPU	4-core	8-core
Memory	8 GB	16 GB or higher
Storage	20 GB HDD	40 GB SSD or higher

Software Requirements

Hardware

Hardware	Minimum	Recommended
CPU	4-core	8-core
Memory	8 GB	16 GB or higher
Storage	20 GB HDD	40 GB SSD or higher

Software

Hardware	Minimum	Recommended
Git	Version control	Latest
Node.js	Backend + tooling	LTS 20.x (18+ works)
npm	Package manager	10+ (bundled with Node LTS)
Python	AI module	3.11+
pip	Python packages	Latest
PostgreSQL client	DB CLI (using Supabase/remote)	14+
Redis	Cache (local or cloud)	7+
VS Code	IDE (recommended)	Latest
Docker	Optional containerized dev	Latest

Major Sofware Packages

Package	Purpose	Version
React	Frontend UI framework	~19.1.0
TypeScript	Type-safe development	~18.x
Jest	Unit testing	^29.7.0
ESLint	Code linting and style enforcement	~8.57.0
Express	API framework	Latest
FastAPI	API framework	>= 0.110
Uvicorn	ASGI server for FastAPI	>= 0.29.0
Transformers	NLP models (Hugging Face)	Latest
Pandas	Data manipulation	>= 2.2.0
Scikit-learn	ML utilities	Latest
Python-dotenv	Environment variable management	Latest
Pytest	Automated testing	>= 7.4.0
Three	Core WebGl-based 3D rendering engine	^0.164.0
React-three	React renderer for Three.js	^8.0.27

Services

This section outlines the external services and APIs required for full functionality of the Gamified Finance application. Services marked with an asterisk (*) are essential for core features such as financial insight generation, user authentication, and cloud-based communication.

Service	Purpose	Required Credentials
Hugging Face	Classification	API key via HF_TOKEN
Supabase PostgreSQL	Primary relational database	Connected via DATABASE_URL
Paseto Token	Stateless authentication using v2 local tokens	key injected via PASETO_LOCAL_KEY
Redis (Railway Proxy)	Session caching and temporary token storage	Connected via REDIS_URL
CoinMarketCap API	Real-time crypto pricing	Accessed via CMC_APIKEY; used in frontend visualizer and backend triggers
Node.js	RESTful API layer	Commnicates with backed serivce
Azure	Deployment service	

Service Dependencies & Secrets

Common ports: Frontend **3000**, API **5000**, AI **6000**, Redis **6379**.
Create a root **.env** (or copy **.env.example**) with at least:

```
# ——— App ———————————————————————————————
NODE_ENV=development
API_PORT=5000
AI_PORT=6000
FRONTEND_PORT=3000

# ——— External Services ——————————————————
DATABASE_URL=postgresql://<user>:<pass>@<host>:5432/<db>
REDIS_URL=redis://<user>:<pass>@<host>:6379

HF_TOKEN=your_hf_token
CMC_APIKEY=your_cmc_api_key
PASETO_LOCAL_KEY=base64_or_hex_secret

# ——— Frontend config ————————————————————
VITE_API_URL=http://localhost:5000/api
VITE_AI_URL=http://localhost:6000
```

Quick Start (All OS)

```
# 1) Clone
git clone https://github.com/COS301-SE-2025/Gamified-Financial-Visualizer.git
cd Gamified-Financial-Visualizer

# 2) Env
cp .env.example .env  # edit values

# 3) Install (root + apps)
npm install --legacy-peers-deps
npm install <package-name> --legacy-peers-deps
npm install --save-dev @types/<package-name>

# 4) Start dev (concurrently runs API, AI, and frontend)
npm run dev
```

Linux Installation Guide

This section provides step-by-step instructions for installing and running the Gamified Finance Visualizer on a Linux system (Ubuntu 20.04+ recommended). Ensure you have root or sudo access.

Prerequisites

- Node.js (v18+)
- Python (v3.10+)
- PostgreSQL client
- Redis client
- Git
- Docker (optional, for containerized setup)

Install core tools:

```
sudo apt update && sudo apt install -y \
nodejs npm python3 python3-pip python3-venv \
postgresql-client redis git
```

Clone the repo

```
git clone https://github.com/COS301-SE-2025/Gamified-Financial-Visualizer.git
cd gamified-finance-visualizer
```

Backend Setup

```
npm install --legacy-peers
cp .env.example .env
cd api-backend
npm install --legacy-peers
npm run build
```

AI Module Setup

```
cd api-backend/modules/ai/app
python3 -m venv venv
pip install - requirements.txt
```

Frontend Setup

```
cd frontend
npm install --legacy-peers
npm run build
```

Run All service

```
npm run dev
```

MacOS installation Guide

This guide walks through installing and running the Gamified Finance Visualizer on macOS. It covers backend (Node.js), AI module (Python), frontend (React), and service dependencies (Supabase, Redis, external APIs).

Prerequisites

- Install core tools via https://brew.sh/
- Python (v3.10+)
- PostgreSQL client
- Redis client
- Git

Docker (optional, for containerized setup)

```
/bin/bash -c "$(curl -fsSL
https://raw.githubusercontent.com/Homebrew/install/HEAD/install.sh)"
brew install node python@3.10 redis git postgresql
```

Tools

```
sudo apt update && sudo apt install -y \
  nodejs npm python3 python3-pip python3-venv \
  postgresql-client redis git
```

Clone the repo

```
git clone https://github.com/COS301-SE-2025/Gamified-Financial-Visualizer.git
cd Gamified-Financial-Visualizer
cp .env.example .env && nano .env
```

Backend Setup

```
npm install --legacy-peers
cp .env.example .env
cd api-backend
npm install --legacy-peers
npm run build
npm run dev   # or from root: npm run dev:api
```

AI Module Setup

```
cd modules/ai/app
python3 -m venv venv
pip install - requirements.txt
python -m uvicorn app.server:app --host 0.0.0.0 --port 6000
```

Docker installation Guide

Frontend Setup
cd frontend
npm install --legacy-peers
npm run dev   # dev server on http://localhost:3000
 # for production build:
npm run build

Run All service
npm run dev

Docker installation Guide

This guide walks through installing and running the Gamified Finance Visualizer using Docker. It covers the full stack: backend (Node.js), AI module (Python), frontend (React), and service dependencies including Supabase, Redis, and external APIs.

Prerequisites

- Docker Engine (v20.10+): Install Docker
- Docker Compose (v2.0+): included with Docker Desktop or install separately
- Git: for cloning the repository

Clone the repo

```
git clone https://github.com/COS301-SE-2025/Gamified-Financial-Visualizer.git
cd Gamified-Financial-Visualizer
cp .env.example .env && nano .env
```

Start All Services

```
docker-compose up --build
```

This spins up:

- backend: Node.js API
- frontend: React dashboard
- ai-module: Python service
- redis: caching layer

Final Comments

Your terminal should look something like this to confirm setup after running the application

```
[3] Issues checking in progress...
[3] No issues found.
[2] INFO:     Started server process [2907]
[2] INFO:     Waiting for application startup.
[2] INFO:     Application startup complete.
[2] INFO:     Uvicorn running on http://0.0.0.0:6000 (Press CTRL+C to quit)
[1] [2025-08-17T00:50:46.401Z] INFO: Auth module registered
[1] [2025-08-17T00:50:46.403Z] INFO: Event listener registered for: transaction.created
[1] [2025-08-17T00:50:46.403Z] INFO: Budget module registered
[1] [2025-08-17T00:50:46.403Z] INFO: Event listener registered for: transaction.created
[1] [2025-08-17T00:50:46.403Z] INFO: Goal module registered
[1] [2025-08-17T00:50:46.403Z] INFO: Learning module registered
[1] [2025-08-17T00:50:46.403Z] INFO: Classifier module registered
[1] [2025-08-17T00:50:46.404Z] INFO: Insights module registered
[1] [2025-08-17T00:50:46.404Z] INFO: Community module registered
[1] [2025-08-17T00:50:46.404Z] INFO: Event listener registered for: transaction.created
[1] [2025-08-17T00:50:46.404Z] INFO: Achievement module registered
[1] [2025-08-17T00:50:46.404Z] INFO: Notifications module registered
[1] [2025-08-17T00:50:46.406Z] INFO: Monolith listening on port 5000 (with Socket.IO)
[1] [2025-08-17T00:50:47.669Z] INFO: Redis ping:
```

Coding standards

To contribute to Gamified Finance, it is suggested to follow the coding standards stipulated by the CodeBlooded team to uphold maintaiability of the application.

Using the Gamified Finance application

To see how to use the appication, refer to our User Manual

Gamified Finance

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